# Candidate 03: the pipeline

DINO gives Z. We transform Z before comparing images.



Image

→ **Frozen DINO** → **Z 384 numbers** → **Shared transform** → **t 256 numbers** → **Cosine and ranking**

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**Our Nine Steps**

1–4: same DINO model and data

5, 6, 8: gallery matrix, SVD, 256 coordinates, soft weights

7, 9: cosine ranking and mAP

# 1. Build the shared transformation

Normalize each gallery Z to y (length 1). Their average is μ.





**4,993 gallery images**

$$\begin{bmatrix} (y_1 - \mu)^T \\ (y_2 - \mu)^T \\ \vdots \\ (y_{4993} - \mu)^T \end{bmatrix} = Y$$

One row per image: 4,993 × 384

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$$Y = U \Sigma V^T$$

Save μ, $V_{256}$ and soft weights for every future comparison.

Fit once from the gallery. No query images or class labels.

# 2. Center and reduce each vector

For any query or gallery image, Z contains 384 numbers.

| Normalize | Center | Project |
| --- | --- | --- |
| $y = Z / \|Z\|$ | $y - \mu$ | $a = V_{256}^{T} (y - \mu)$ |
| Length becomes 1 | Subtract the same μ | Use the same 256 directions |

384 coordinates → 256 new coordinates

$V_{256}$ keeps directions paired with the 256 largest singular values.

# 3. Reweight, normalize, compare

Soft whitening gives each coordinate its own weight.

Two-coordinate illustration

| Coordinates | | Relative weights | | Weighted vector | | Length 1 |
|---|---|---|---|---|---|---|
| $(0.2, 0.1)$ | ➡ | $\times (0.5, 1)$ | ➡ | $(0.1, 0.1)$ | ➡ | $(1/\sqrt{2}, 1/\sqrt{2})$ |

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Query and gallery use the same transformation.

$$\text{score} = t_Q^{\mathrm{T}} t_I$$

Higher cosine gives a higher rank.

The weights depend on variation across the gallery.

# Ten experiments

10 predefined configurations. All 10 completed.

| ID | Input | Output size | Weights |
|---|---|---|---|
| 01 | CLS | 384 | None |
| 02 | CLS | 384 | Soft whitening |
| **03** | **CLS** | **256** | **Soft whitening** |
| 04 | CLS | 128 | Soft whitening |
| 05 | CLS | 256 | Full whitening |
| 06 | CLS | 128 | Full whitening |
| 07 | CLS + mean patch | 256 | Soft whitening |
| 08 | CLS + mean patch | 128 | Soft whitening |
| 09 | CLS + mean patch | 256 | Full whitening |
| 10 | CLS + mean patch | 128 | Full whitening |

All subtract the gallery average. Mean patch = the average patch vector.

# Oxford result

70 queries. 4,993 gallery images. The same frozen DINO.

| Method | Medium mAP (%) | Hard mAP (%) |
|---|---|---|
| Local CLS reference | 32.25 | 7.82 |
| **Candidate 03** | **34.48** | **8.73** |

**+2.23 percentage points** **+0.91 percentage points**

4 of the 10 methods improved both Oxford scores.

Candidate 03 has not been tested on Paris.

Saved results and all ten configurations